Derivatives of Inverse Trig

Section 5.6
1-6-26

A) Review

$\sin^{-1}(x) \neq \frac{1}{\sin(x)}$

$\sin^{-1}(x) = \arcsin(x)$

Ex: $\sin x = \frac{1}{2}$

$x = \frac{\pi}{6}, \frac{5\pi}{6}$

Ex: $\arcsin\left(\frac{1}{2}\right) = \frac{\pi}{6}$

Ex: $\arcsin\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$

Ex: $\cos(\arcsin\frac{2}{3})$

$= \frac{\sqrt{5}}{3}$

Ex: $\cos(\arcsin x)$

$= \frac{\sqrt{1-x^2}}{1} = \sqrt{1-x^2}$

B) Derivatives

Ex: $y = \arcsin x$

$\sin y = x$

$\cos y \frac{dy}{dx} = 1$

$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\cos(\arcsin x)} = \frac{1}{\sqrt{1-x^2}}$

ARCSIN rule!

$$\frac{d}{dv}\arcsin v = \frac{v'}{\sqrt{1-v^2}}$$

Ex: $\arcsin(3x) = \frac{3}{\sqrt{1-9x^2}}$

Ex: $\arccos x = y$

$\cos y = x$

$-\sin y = 1$

ARCCOS rule!

$$\frac{d}{dv}\arccos v = -\frac{v'}{\sqrt{1-v^2}}$$

Ex: $y = \arctan x$

$\tan y = x$

$\sec^2 y \frac{dy}{dx} = 1$

$\frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{\sec^2(\arctan x)} = \frac{1}{1+x^2}$

ARCTAN rule!

$$\frac{d}{dv}\arctan v = \frac{v'}{1+v^2}$$

ARCCOT rule!

$$\frac{d}{dv}\operatorname{arccot} v = -\frac{v'}{1+v^2}$$

Ex: $y = \operatorname{arcsec} x$

$\sec y = x$

$\sec y \tan y \frac{dy}{dx} = 1$

$\frac{dy}{dx} = \frac{1}{\sec y \tan y} = \frac{1}{\sec(\operatorname{arcsec} x)\tan(\operatorname{arcsec} x)} = \frac{1}{|x|\sqrt{x^2-1}}$

ARCSEC rule!

$$\frac{d}{dv}\operatorname{arcsec} v = \frac{v'}{|v|\sqrt{v^2-1}}$$

ARC CSC rule!

$$\frac{d}{dv}\operatorname{arccsc} v = -\frac{v'}{|v|\sqrt{v^2-1}}$$

Ex: $x \cdot \arctan(x^2) =$

$\frac{3x^2}{1+x^6} + \arctan(x^2)$

$\frac{3x^2}{1+x^6} + \arctan(x^2)$

Ex: $\operatorname{arcsec}(e^x) =$

$= \frac{e^x}{|e^x|\sqrt{e^{2x}-1}}$

$= \frac{1}{\sqrt{e^{2x}-1}}$

ALL Inv. Trig Deriv. Rules

ARCSIN

$$\frac{d}{dv}\arcsin v = \frac{v'}{\sqrt{1-v^2}}$$

ARCCOS

$$\frac{d}{dv}\arccos v = -\frac{v'}{\sqrt{1-v^2}}$$

ARCTAN

$$\frac{d}{dv}\arctan v = \frac{v'}{1+v^2}$$

ARCCOT

$$\frac{d}{dv}\operatorname{arccot} v = -\frac{v'}{1+v^2}$$

ARCSEC

$$\frac{d}{dv}\operatorname{arcsec} v = \frac{v'}{|v|\sqrt{v^2-1}}$$

ARCCSC

$$\frac{d}{dv}\operatorname{arccsc} v = -\frac{v'}{|v|\sqrt{v^2-1}}$$